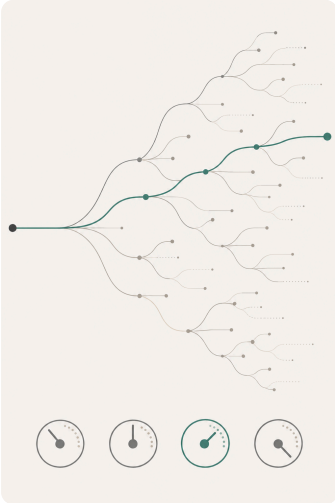
What structurally breaks an agent?

Research proposal - 1 Nov 2026

A plan to measure which structural properties of a task predict agent failure, by systematically varying task structure and testing which factors predict failure under controlled conditions.



Editorial illustration, generated for this proposal. Not data.

The idea

Agent benchmarks return one score per model. When a model fails, the score cannot say why: the number of dependent steps, the choices available at each step, the state it had to remember, or what the environment hid. Those properties all covary in realistic tasks, so failure analysis stays anecdotal.

This proposal makes task structure the experimental variable. A generator emits small executable tool-use workflows over a hidden state machine, where each structural dimension is set directly. Instances are kept small with an explicit finite horizon, so structural metrics and solvability can be computed exactly and, where feasible, validated against an exhaustive solver. That validates the manipulation, not the model: structural metrics and optimal solvability describe the task, and say nothing by themselves about how hard it is for an LLM. Model difficulty is an empirical outcome of the study, not a property assumed at design time.

Your VISTA-Bench manuscript reported interaction depth and branching factor as descriptive statistics of its environments. This study takes the natural next step: it treats structure itself as the manipulated variable, in minimal tasks, with no app synthesis and no knowledge-conflict angle.

The four dimensions

1. **Dependency depth**
   How many steps must happen in a strict order before the goal is reachable.

2. **Branching factor**
   How many plausible actions exist at each step, so success depends on rejecting invalid moves.

3. **State width**
   How many internal variables must be tracked to choose the next action correctly.

4. **Observability**
   How much of that hidden state the environment's feedback reveals after each action.

Coupling, estimands and controls

These dimensions are not naturally independent. More depth at fixed branching multiplies the number of reachable paths; branching changes the entropy of each action choice; low observability only bites when there is state worth tracking. The design treats that coupling explicitly instead of pretending the knobs are orthogonal.

1. Estimands. The primary estimand for each dimension is its main effect on task success across a factorial grid: the average success-rate change between its levels, averaged over every combination of the other dimensions, within one generator family. Interactions between dimensions are estimated from the same grid and reported in full, not assumed away.

2. Manipulation checks. Every generated task is scored with graph metrics computed directly from its state machine: critical-path length, mean out-degree, reachable-state count, and observation-state mutual information. A task is admitted only if moving its target knob moved the target metric while keeping the other metrics within tolerance; otherwise it is regenerated.

3. Matched controls. Surface confounds are fixed by construction: same vocabulary, near-equal prompt and feedback token counts, same number of exposed tools. Where the grid allows it, a dedicated control holds total reachable states constant while one knob varies, so an observed effect cannot be dismissed as just a bigger search space. Because exact matching is infeasible for some grid cells, feasibility is checked per cell, non-matched cells are dropped from that control, and its coverage is reported rather than assumed.

4. Scope of the causal claim. Claims are causal only within the generator family and only under the stated invariance assumptions: content, wording and reward structure held fixed. The output is a measurement of structural sensitivity, not a universal law of agent failure. Replication in a second, independently designed generator family is a planned robustness check.

What the experiment returns

- Per-dimension scaling curves: task success against each dimension's level, for several frontier models.
- A failure threshold per dimension per model: a structural profile of an agent instead of a single score.
- A reasoning arm: whether extra test-time compute shifts the thresholds, and whether it shifts them equally on every dimension, against a token-budget-matched control.

Bounded claims

This is a proposal. No tasks have been generated and no models have been run. Score differences on existing benchmarks motivated the question but say nothing about causes; only controlled manipulation can, and only within the limits stated above.

Closest work and the gap

1. **The Illusion of Thinking (Apple, 2025)**
   Varies problem complexity in non-interactive puzzles. Interactive, stateful tool use is a different setting.

2. **METR: Measuring AI Ability to Complete Long Software Tasks (2025)**
   Time horizons on real tasks, where all structural properties covary at once. This design separates them.

3. **ARIES (2025)**
   Interactive thought-graph environments for abstract reasoning, not stateful tool-use workflows.

4. **Task complexity in multi-agent evaluation (2025)**
   Argues complexity matters for multi-agent systems; does not isolate per-dimension effects for single tool-using agents.

5. **tau-bench (2024)**
   A stateful tool-agent-user benchmark with fixed difficulty; a target for measurement, not a controlled instrument.

The gap this claims is specific and checkable: a factorial, per-dimension isolation of structural difficulty for stateful tool-using agents, with manipulation checks and content held constant. It is not a claim that nobody has studied task difficulty.

arxiv.org/abs/2506.06941 · arxiv.org/abs/2503.14499 · arxiv.org/abs/2502.21208 · arxiv.org/abs/2510.04311 · arxiv.org/abs/2406.12045

Why this fits DeepMind

DeepMind's Frontier Safety Framework ties decisions to measured capability thresholds, and its dangerous-capability evaluations depend on knowing what an evaluation actually measures. Structural profiles make evaluation difficulty auditable, and per-dimension thresholds could sharpen capability forecasting beyond duration-based estimates alone.

deepmind.google/blog/strengthening-our-frontier-safety-framework · deepmind.google/research/publications/78150

From proposal to arXiv

1. Build the generator and verify the manipulation checks on graph metrics.
2. Calibrate knob levels on one model so curves span floor to ceiling.
3. Preregister the factorial grid, estimands and exclusion rules.
4. Run four to six frontier models with matched token budgets.
5. Replicate in a second generator family; report all interactions and negative results.

Proposal prepared 1 Nov 2026. No experiments have been run; every level and budget named here is a design choice, not a result. Related-work links verified 1 Nov 2026.